

績效管理系統 期末報告

題目說明：

以雲原生架構打造績效管理平台，為企業提供即時、透明、可稽核的人才管理解決方案

第11組

| Agenda

- 1 Overview**
- 2 System Architecture**
- 3 Demo: RBAC & Audit Log**
- 4 Testing**
- 5 DevOps**

User Story (1/6)

目標設定與追蹤：支援 SMART 目標設定，可設定個人或團隊目標

[身為]

每天工作很忙碌的高階主管，

[我想要]

可以協助員工設定個人目標並發布明確的 KPI，

[這樣才能]

精確地向員工傳達績效標準，並有效率的維持員工工作重心和企業戰略一致



User Story (2/6)

進度更新：員工可定期更新目標進度，主管可查看相關更新

[身為]

在意績效的大型企業員工，

[我想要]

能夠隨時檢視個人目標，

[這樣才能]

明確知道該做什麼工作才能提升績效



User Story (3/6)

多種評估模板：支援多種評估類型，
可自定義評估問卷和評分標準

[身為]

同時負責多個部門考核的 HR，

[我想要]

能為不同職能的員工套用對應的評估模板，

[這樣才能]

確保各部門考核問題切合實際工作內容，而不是讓所有人填一份問卷導致評估結果流於形式



User Story (4/6)

評估結果發布與確認：評估結果可發布給員工，員工可確認或者提交異議

[身為]

在意績效的大型企業員工，

[我想要]

能夠預覽結果，並向主管針對其中提出問題，

[這樣才能]

感覺績效考核是公正、公開、透明的，並且獲得更明確的回應



User Story (5/6)

基於角色的存取控制：確保只有授權人員才能存取特定績效資料和功能

[身為]

重視資料安全的系統管理員，

[我想要]

系統具備嚴格的基於角色存取控制機制，

[這樣才能]

確保只有經過授權的人員才能存取或修改特定的績效資料，以保障員工隱私與企業的合規性



User Story (6/6)

詳細的審計日誌：紀錄所有操作、資料存取和修改，且日誌不可篡改

[身為]

同時負責多個部門考核的 HR，

[我想要]

查閱所有（含修改）評估的完整操作紀錄，

[這樣才能]

在申訴或爭議發生時，提供有系統依據的操作歷程，而不是靠截圖或 Email 來舉證



Tech Stack

Frontend

- **React + TypeScript**
- **Firebase Hosting**

Backend

- **Java Spring Boot**
- **Google Cloud Run**

Data

- **PostgreSQL**
- **Spring Data JPA**
- **Flyway**

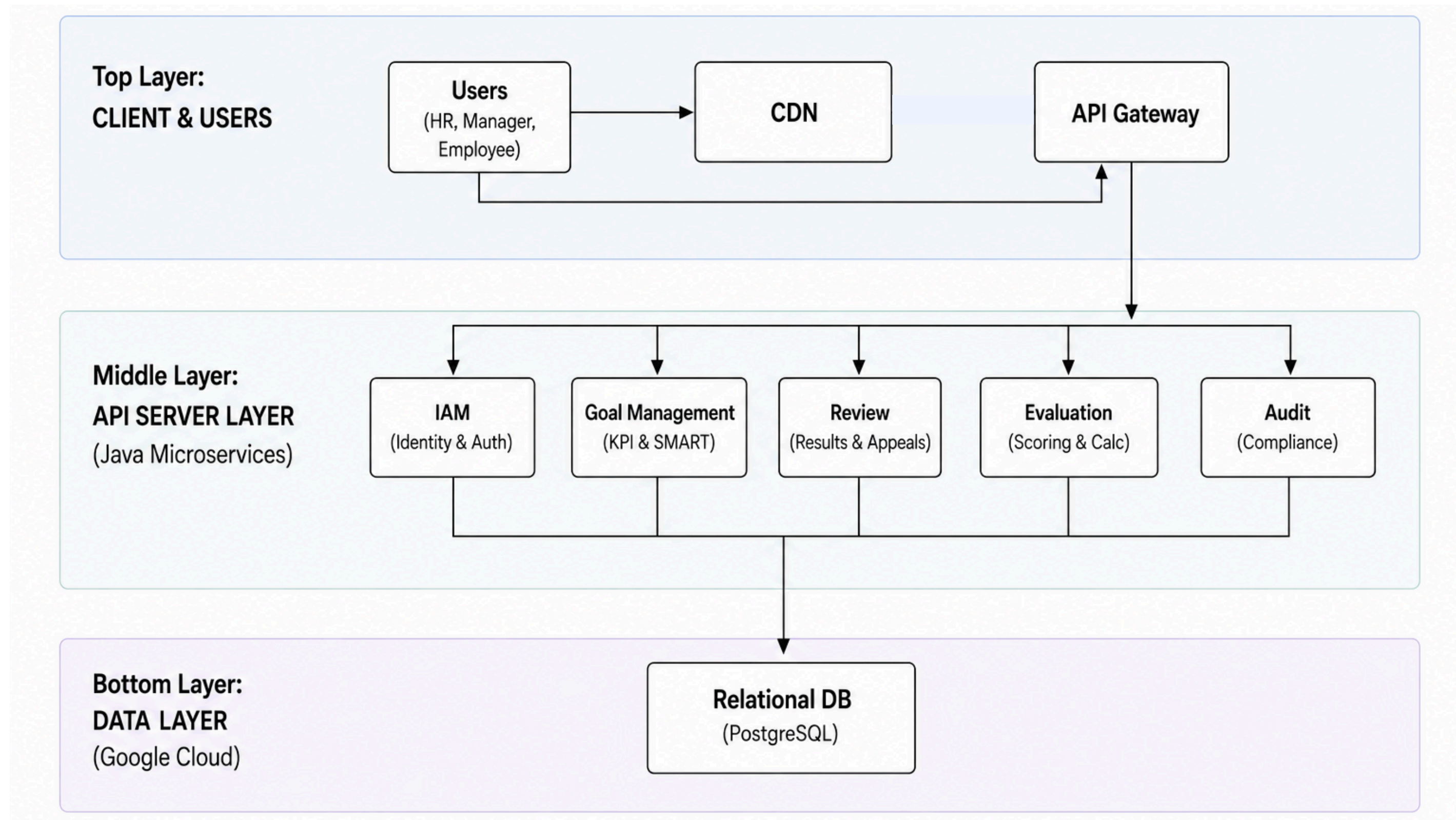
Foundation & DevOps

- **Security: Google SSO (MFA) / AES-256 資料加密 / RBAC 權限控管**
- **CI/CD: GitHub Actions 自動化測試與部署**
- **Reliability: k6 Load Test / Grafana 監控**

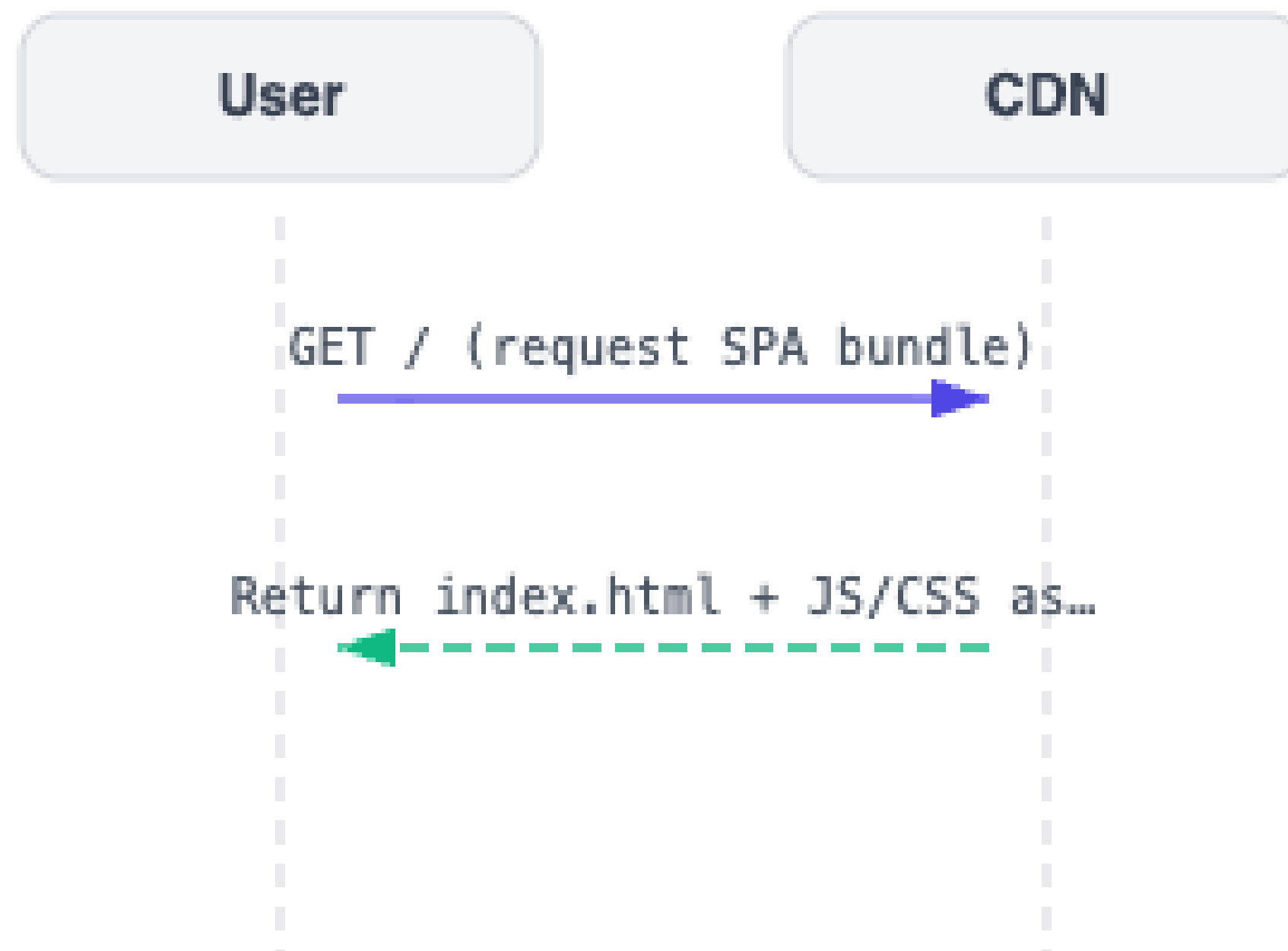
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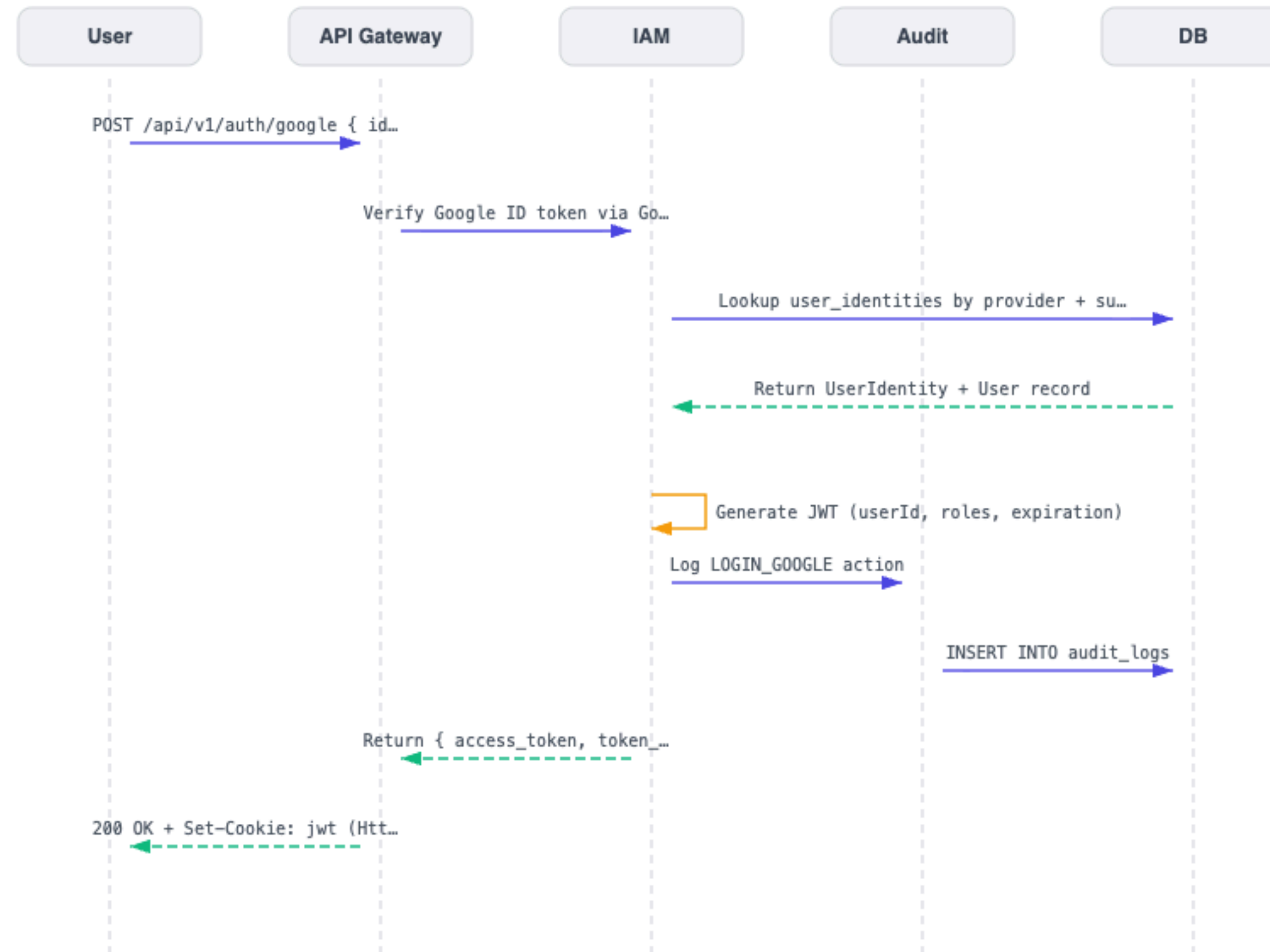
System Architecture



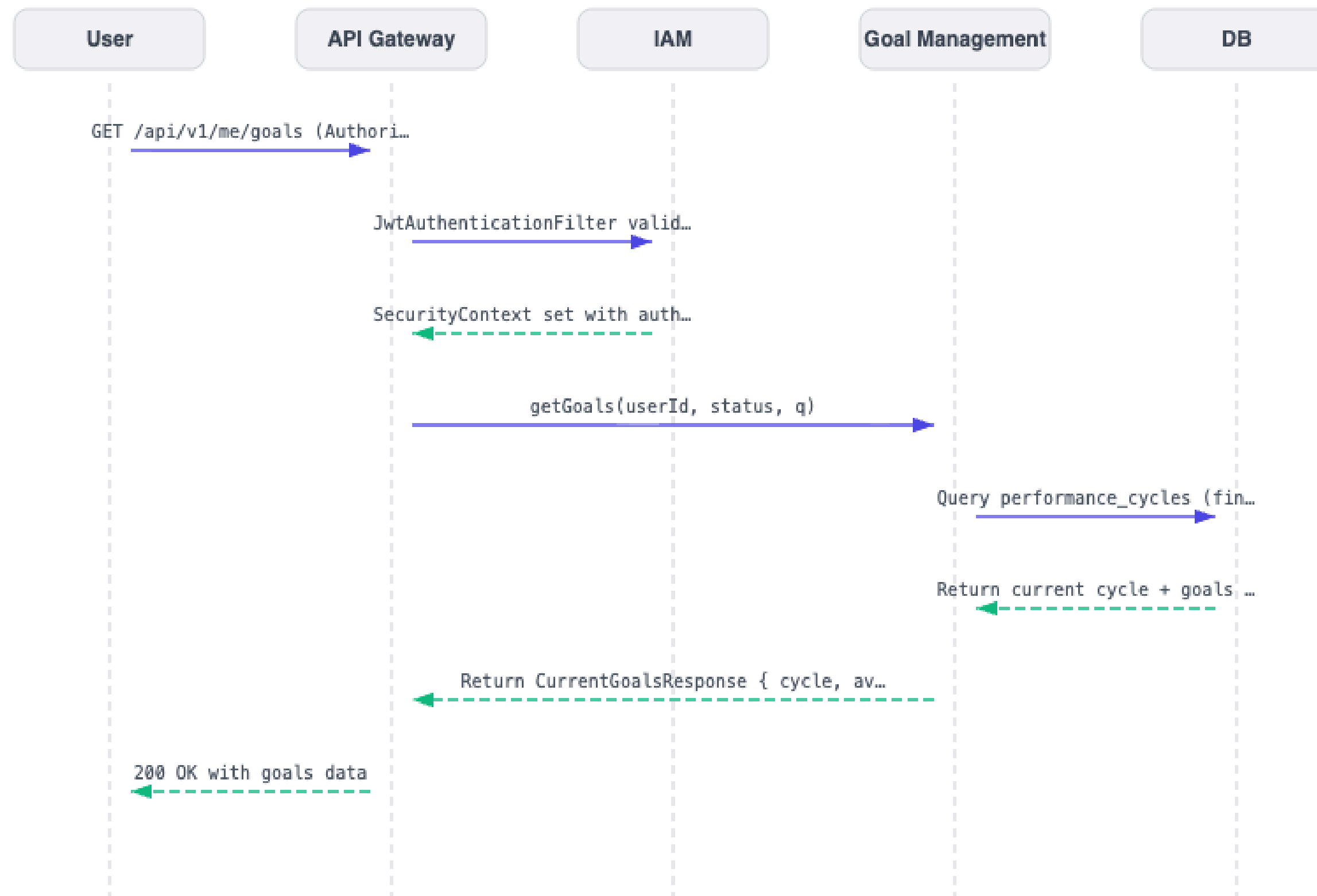
Sequence Diagram (Open Page)



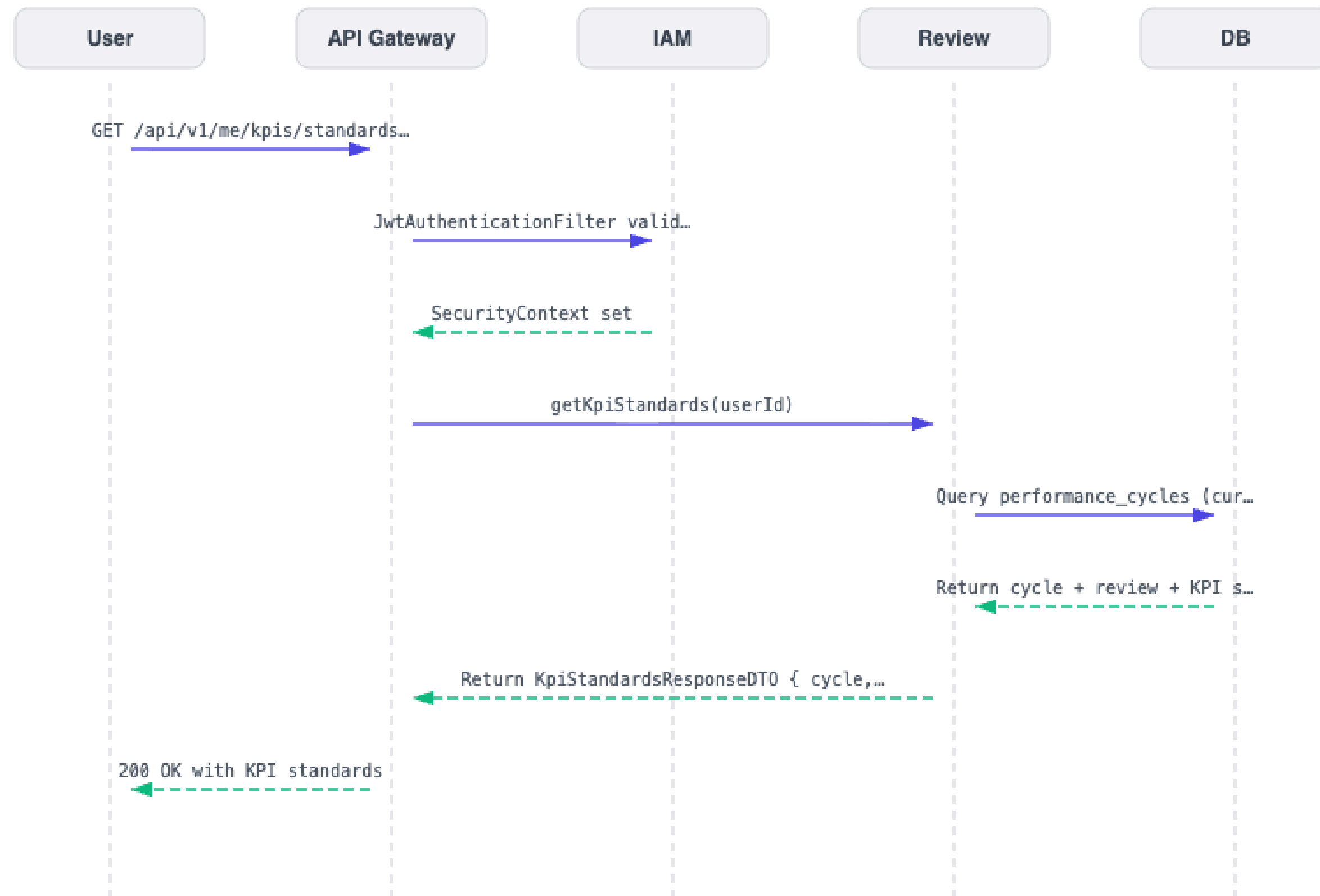
Sequence Diagram (Login)



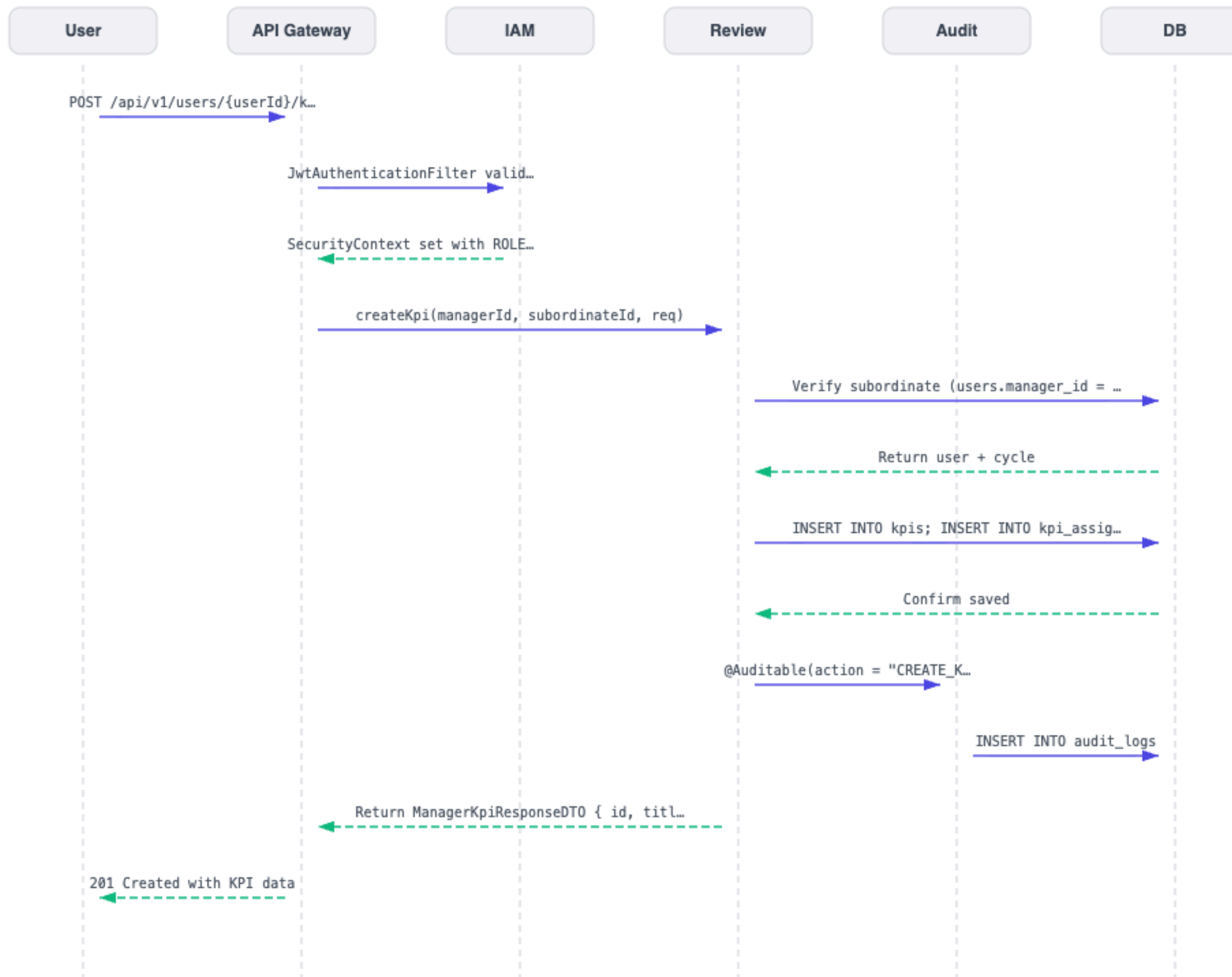
Sequence Diagram (View Goals)



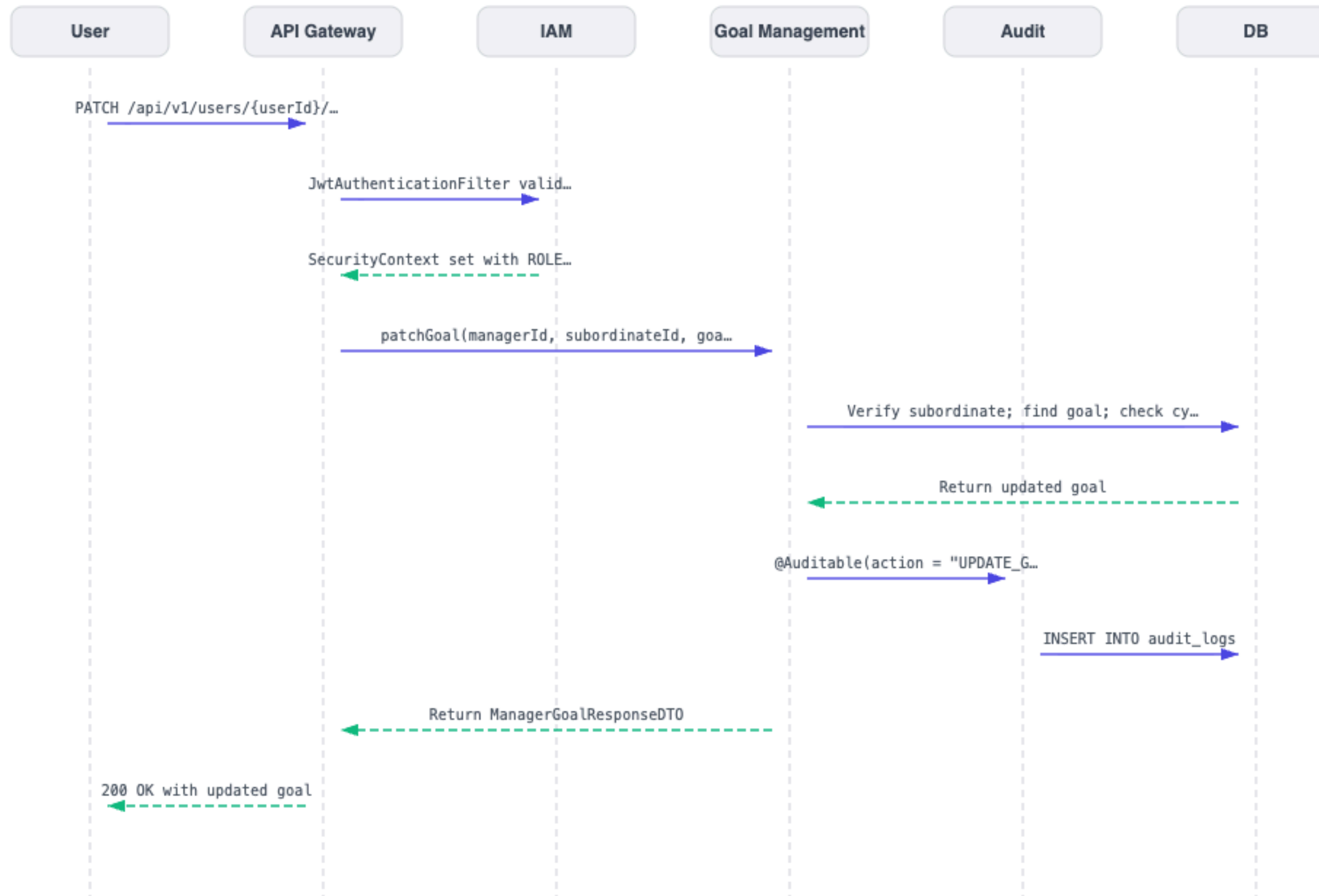
Sequence Diagram (View KPIs)



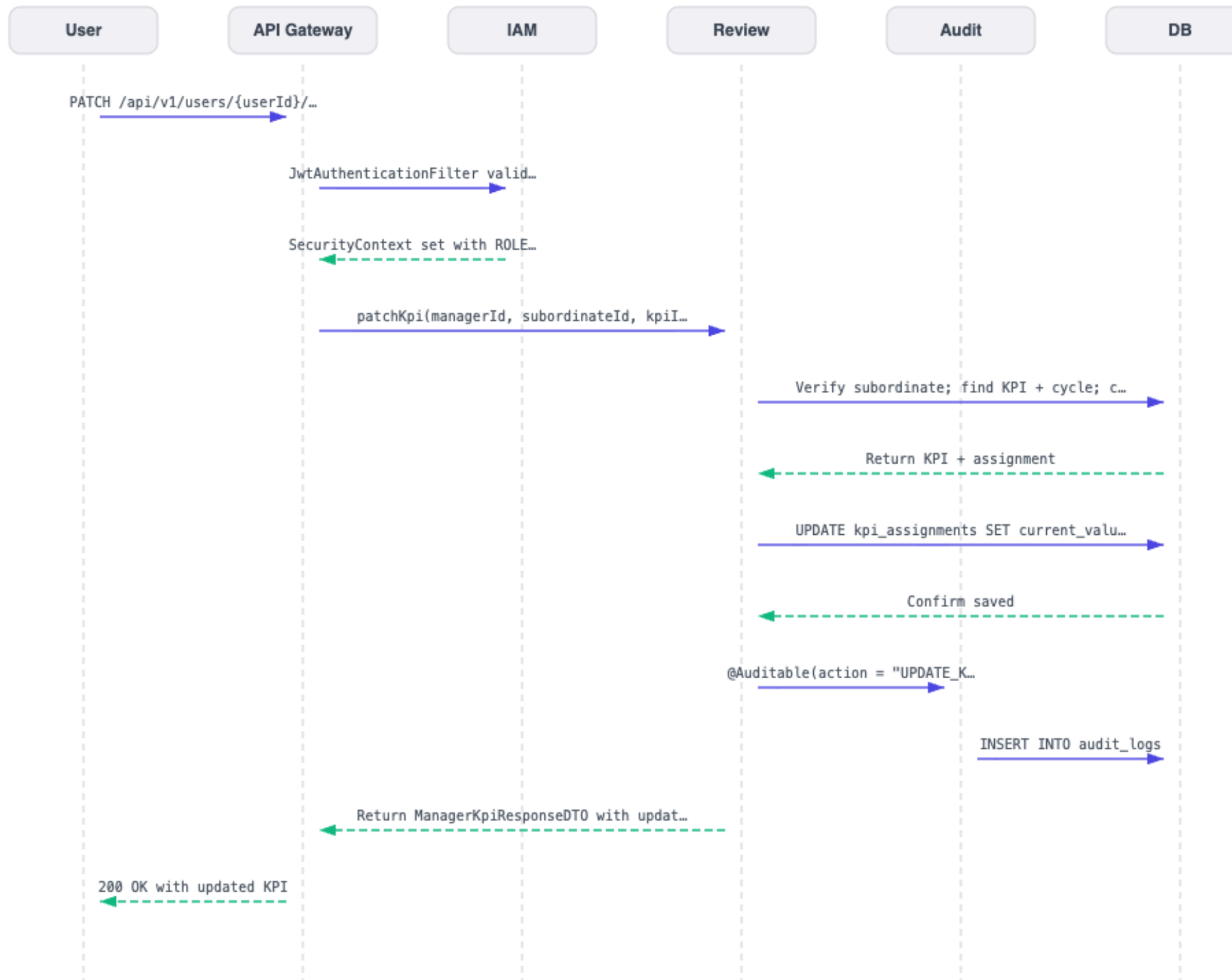
Sequence Diagram (Manager Set KPI Standards)



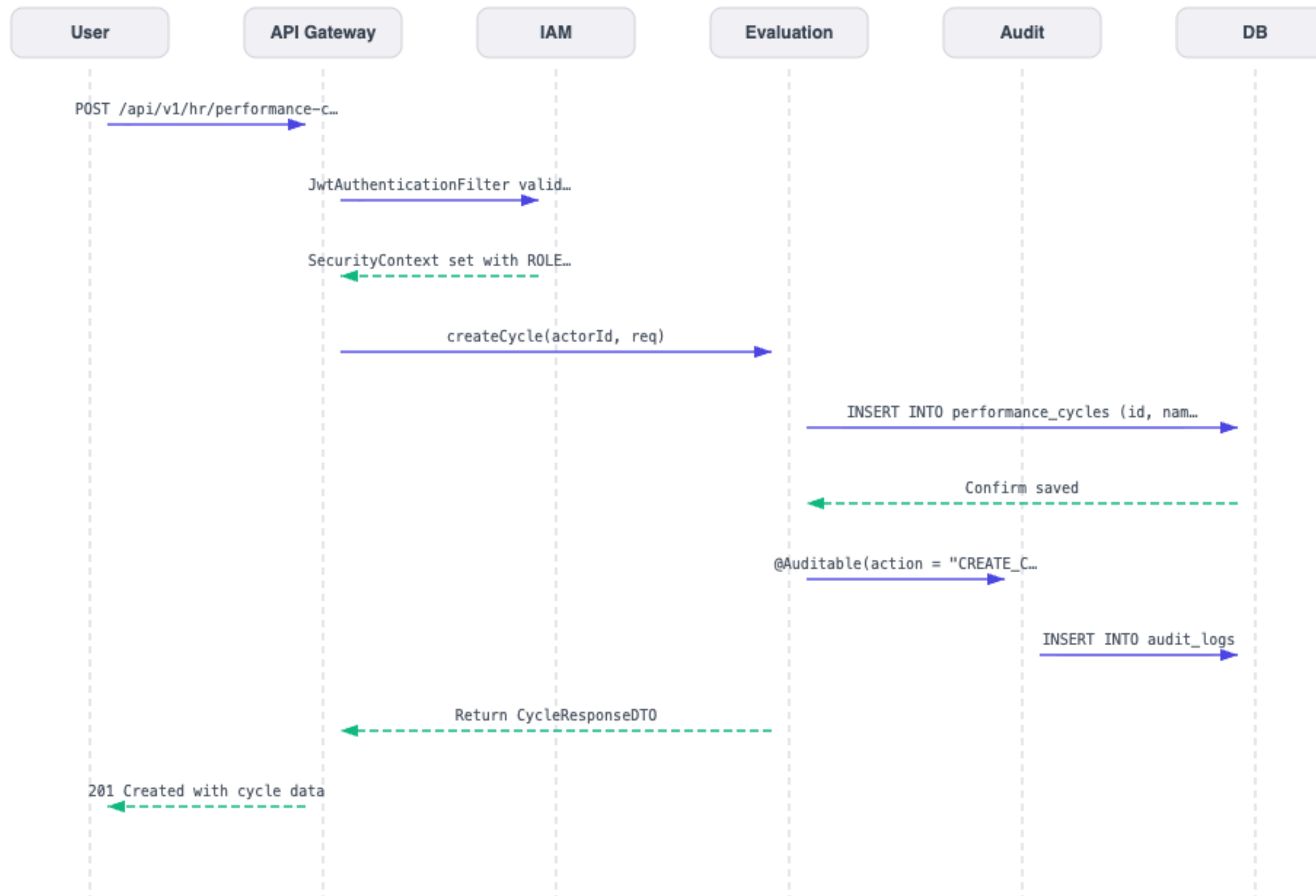
Sequence Diagram (Manager Review Goals)



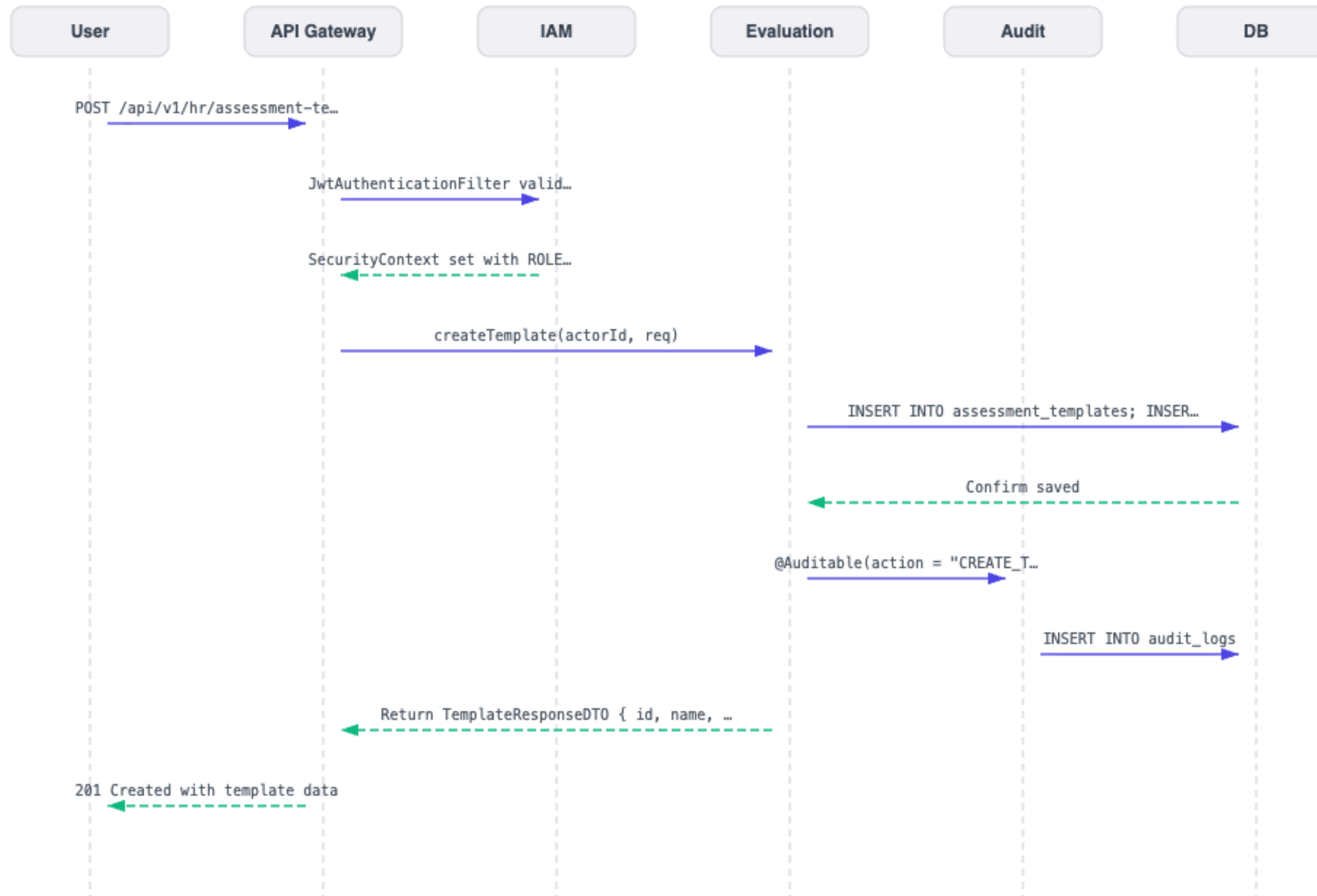
Sequence Diagram (Manager Set KPI Values)



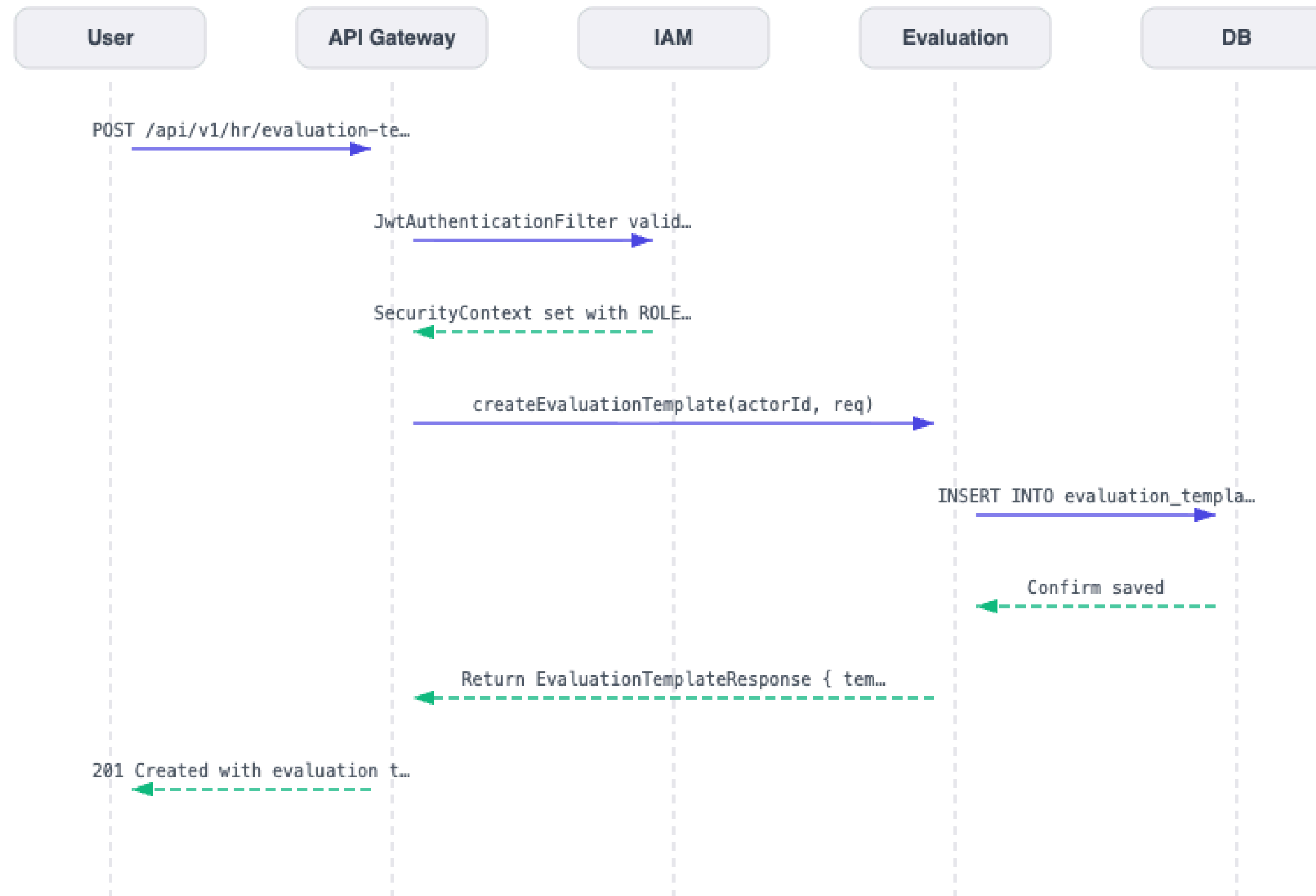
Sequence Diagram (HR Create Cycle)



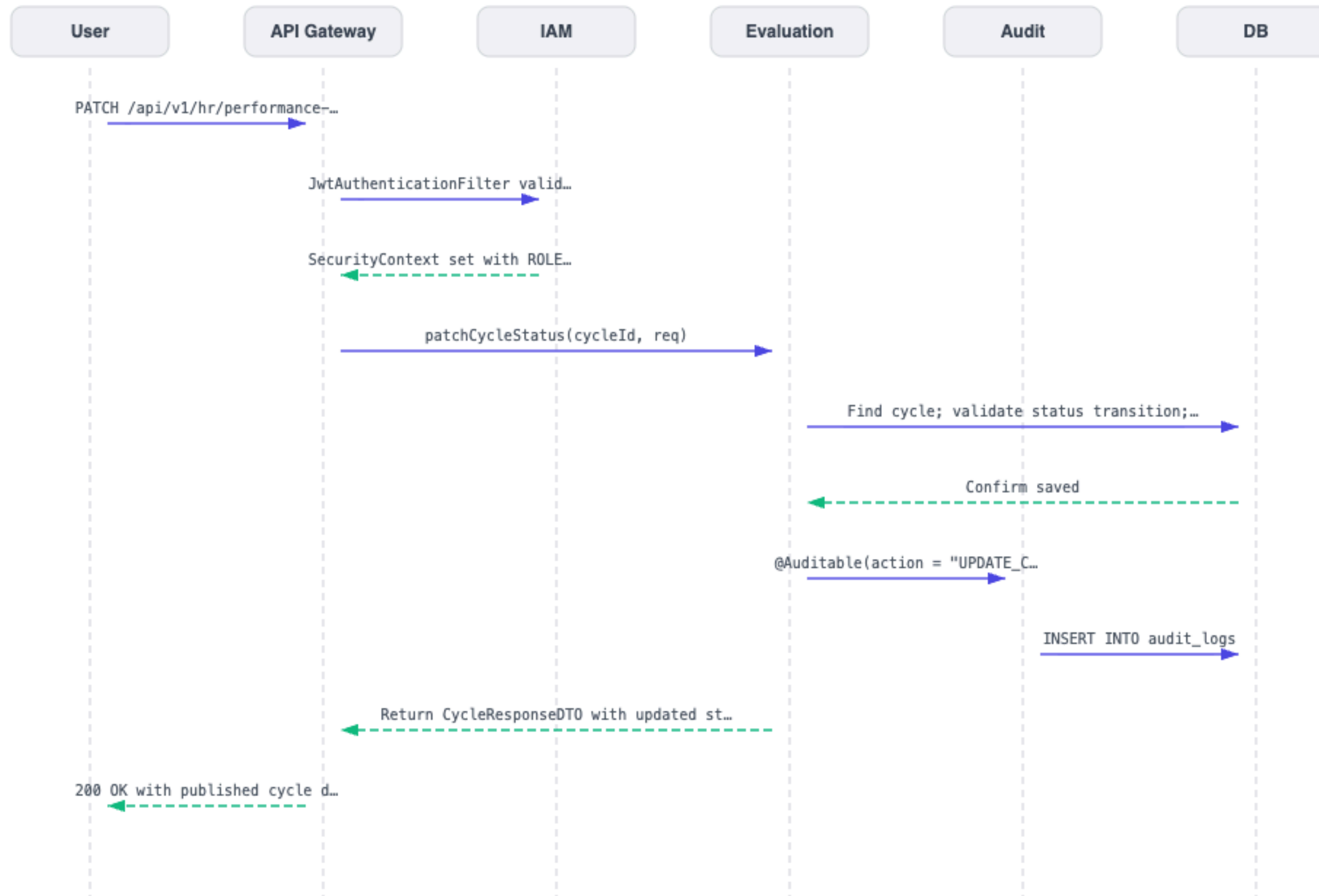
Sequence Diagram (HR Create Questions Template)



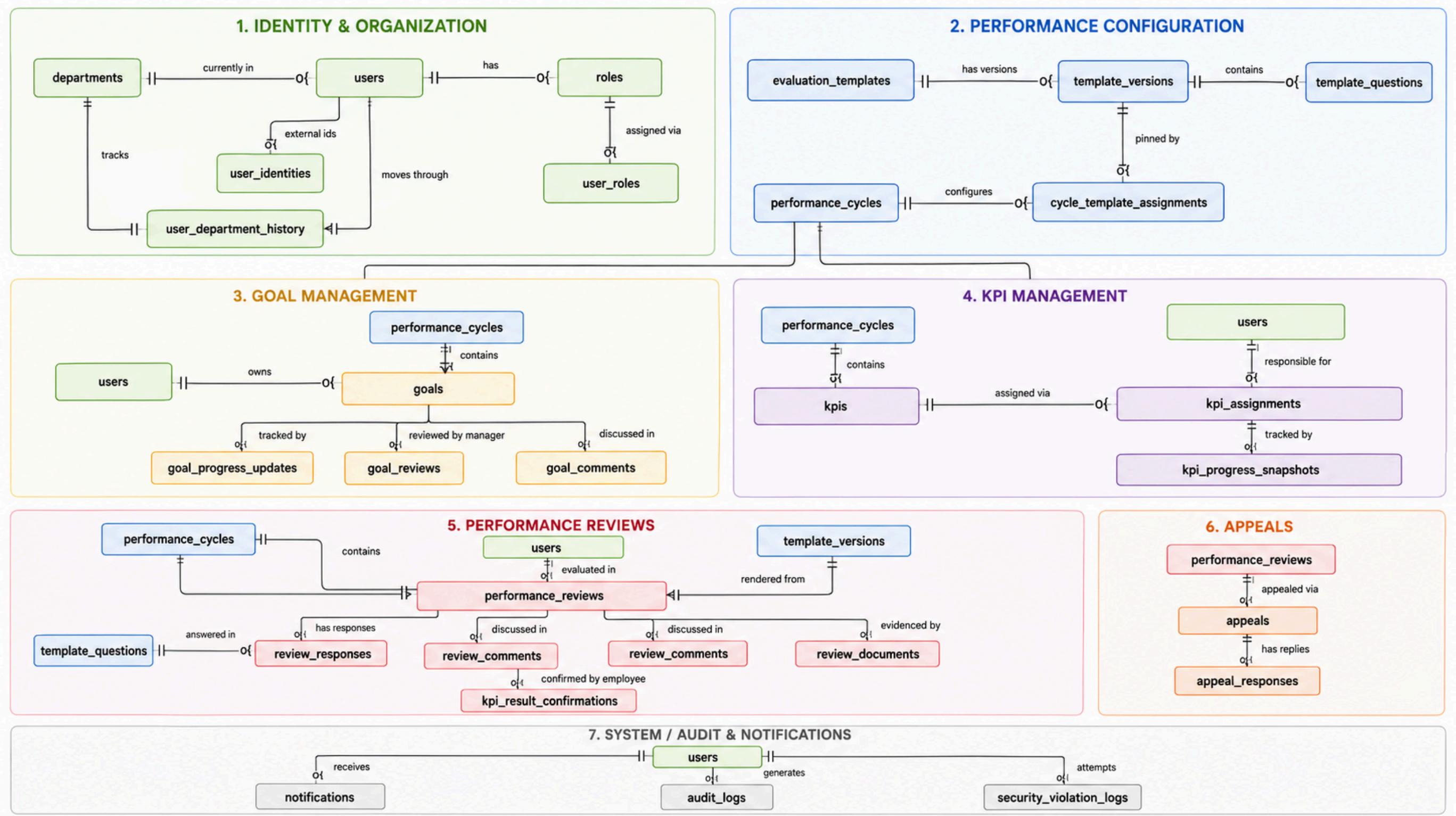
Sequence Diagram (HR Create Evaluation Template)



Sequence Diagram (HR Publish Results)



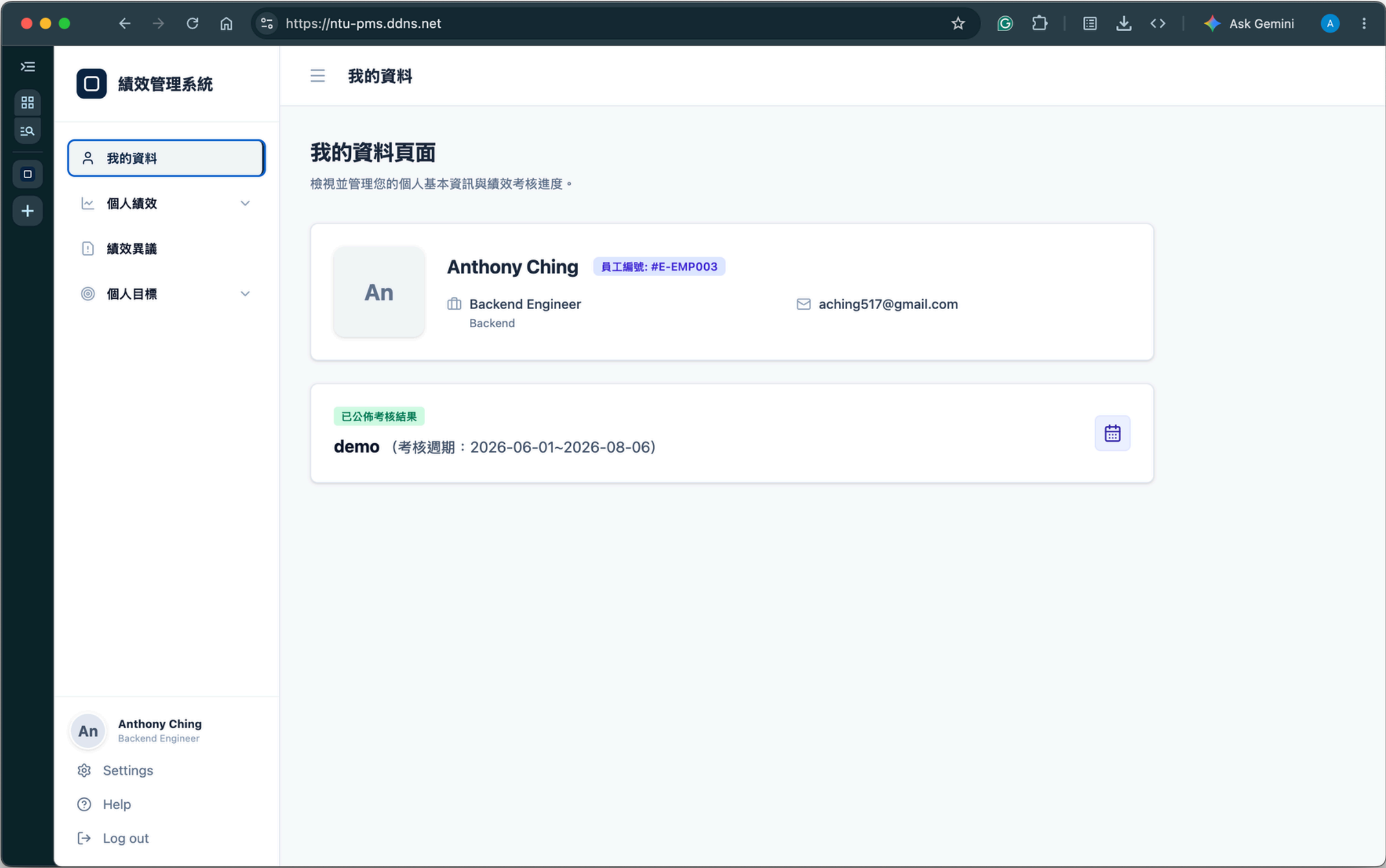
Entity Relationship Diagram



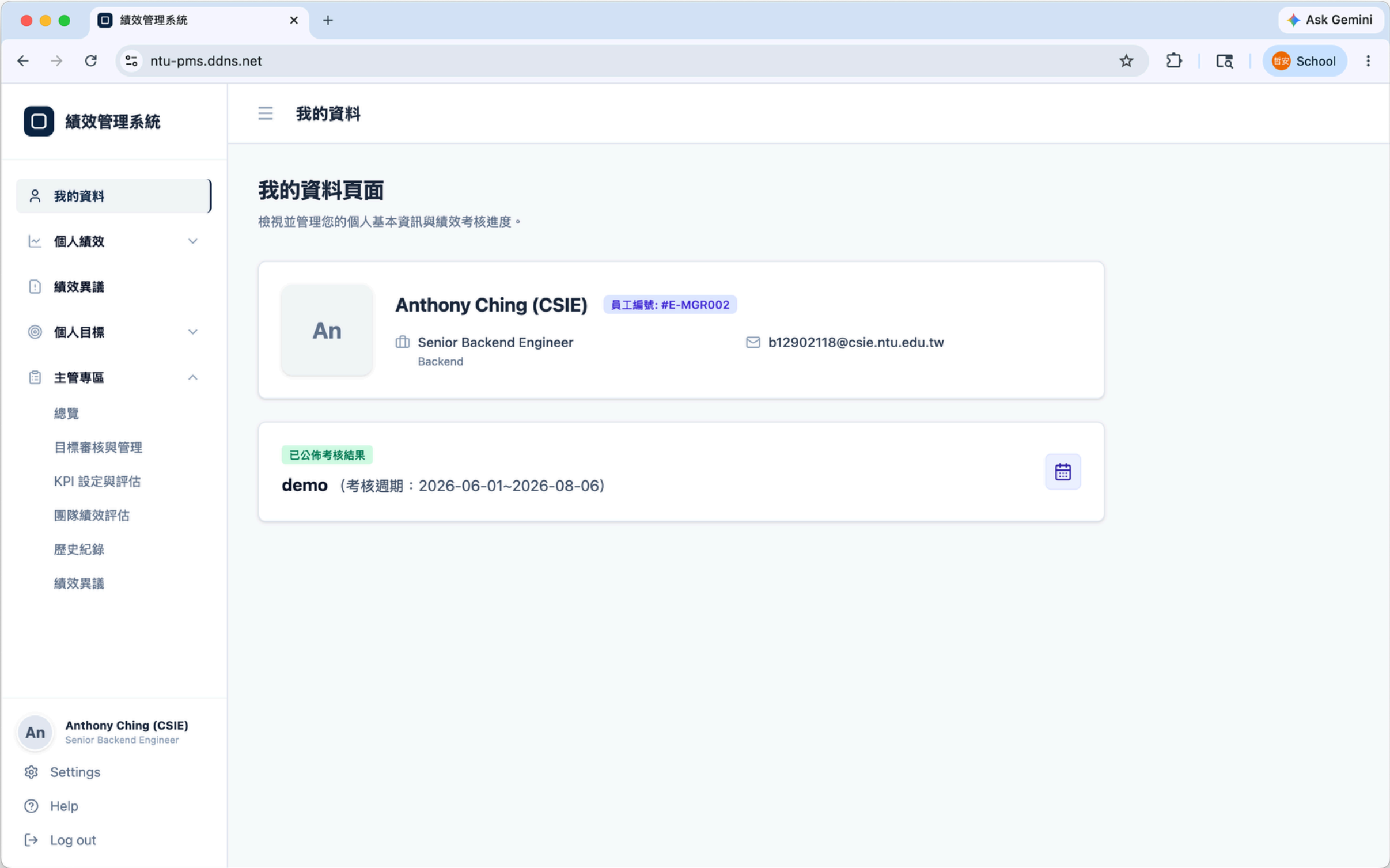
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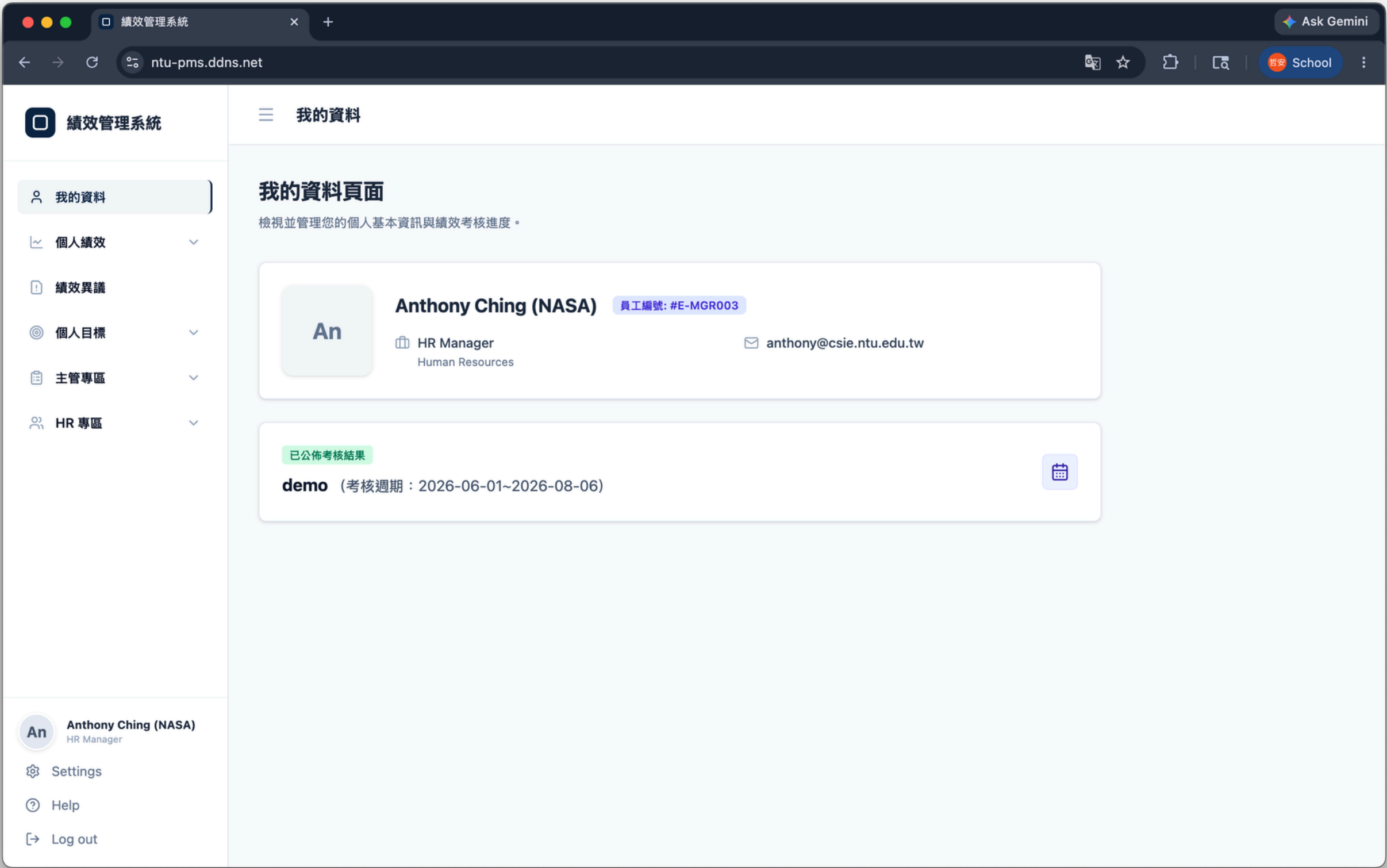
普通員工身份無法瀏覽主管或 HR 專區



主管身份可以瀏覽主管專區



HR (兼主管)身份可以瀏覽 HR (兼主管)專區



Audit Logs

績效管理系統

我的資料

個人績效

績效異議

個人目標

主管專區

HR 專區

考核週期

問卷模板

考核模板

發佈考核結果

查詢稽核紀錄

AnAnthony Ching (NASA)
HR Manager

Settings

Help

Log out

HR 專區 > 查詢稽核紀錄

查詢稽核紀錄

追蹤系統內的所有重要操作與變更，以確保資料安全與考核公正性。

搜尋操作者、動作或資源名稱...

所有類別

操作時間	操作者	事件類別	動作	操作資源 / 內容	來源 IP
2026-06-02 08:05:02	Anthony Ching (NASA)	系統發佈	LOGIN_GOOGLE	auth	-
2026-06-02 07:54:44	Anthony Ching (CSIE)	申覆申訴	RESPOND_TO_APPEAL	appeal	-
2026-06-02 07:54:28	Anthony Ching	申覆申訴	SUBMIT_APPEAL	appeal	-
2026-06-02 07:54:06	Anthony Ching (NTU)	系統發佈	UPDATE_CYCLE_STATUS	performance_cycle	-
2026-06-02 07:53:59	Anthony Ching (NTU)	系統發佈	UPDATE_CYCLE_STATUS	performance_cycle	-
2026-06-02 07:53:49	Anthony Ching (CSIE)	目標指標	SAVE_KPI_EVALUATION	performance_review	-
2026-06-02 07:53:49	Anthony Ching (CSIE)	目標指標	UPDATE_KPI	kpi	-
2026-06-02 07:53:49	Anthony Ching (CSIE)	評分考核	SAVE_QUESTIONNAIRE	performance_review	-

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Code Quality & CI

- **自動化 Pipeline**：每次 Push code，GitHub Actions 皆會自動觸發檢查。
- **Maven 程式碼規範**：透過 Checkstyle、Spotless 統一排版，並使用 SpotBugs 抓出潛在 Bug。
- **安全與覆蓋率**：整合 SonarCloud 進行 Vulnerabilities 與測試覆蓋率檢查。

CI/CD pipeline

AnthonyChing / performance-management-system

Code

Issues5

Pull requests8

Agents

Discussions

Actions

Projects

Wiki

Security and quality

Insights

Settings

Actions

New workflow

All workflows

Backend Pipeline

Copilot

Copilot cloud agent

Dependabot Updates

Frontend Pipeline

Load Test (k6)

Management

Caches

Attestations

Runners

Usage metrics

Performance metrics

All workflows

Showing runs from all workflows

441 workflow runs

WorkflowEventStatusBranchActor

✓ Merge pull request #178 from AnthonyChing/frontend/fix/main...v1.1.2

Frontend Pipeline #91: Commit 492bfb4 pushed by AnthonyChing

10 minutes ago1m 6s

✓ Merge pull request #178 from AnthonyChing/frontend/fix/main...v1.1.2

Backend Pipeline #121: Commit 492bfb4 pushed by AnthonyChing

10 minutes ago5m 31s

✗ Merge pull request #169 from AnthonyChing/fullstack/fix/empl...v1.1.1

Backend Pipeline #120: Commit 5fcd900 pushed by sunSkywalker

19 minutes ago20s

✓ Merge pull request #169 from AnthonyChing/fullstack/fix/empl...v1.1.1

Frontend Pipeline #90: Commit 5fcd900 pushed by sunSkywalker

19 minutes ago1m 6s

✓ chore: fix SonarQube code smells (constants, complexity, injec...backend/chore/sonar-fixes

Backend Pipeline #119: Pull request #177 synchronize by ernestii26

24 minutes ago3m 9s

✓ Merge pull request #178 from AnthonyChing/frontend/fix/main...main

Backend Pipeline #118: Commit 492bfb4 pushed by sunSkywalker

27 minutes ago5m 8s

✓ fix formatfrontend/fix/main-test-fa...

Backend Pipeline #117: Pull request #178 opened by sunSkywalker

30 minutes ago3m 7s

Overview

System Architecture

Demo: RBAC & Audit Log

Testing

DevOps

31

CI/CD pipeline

← Frontend Pipeline

✓ Merge pull request #178 from AnthonyChing/frontend/fix/main-test-fail #91

Re-run all jobs ⋮

Summary

All jobs

✓ frontend-test

✓ frontend-build-and-deploy

Run details

Usage

Workflow file

Triggered via push 11 minutes ago

AnthonyChing pushed

492bfb4

v1.1.2

Status

Success

Total duration

1m 6s

Artifacts

—

frontend.yml

on: push

✓ frontend-test 34s

✓ frontend-build-and-deploy 25s

⌂ − +

frontend-test summary

⋮

Vitest Test Report

Summary

• Test Files: ✓ 15 passes · 15 total

• Test Results: ✓ 108 passes · 108 total

[Job summary generated at run-time](#)

CI/CD pipeline

← Backend Pipeline

✓ Merge pull request #178 from AnthonyChing/frontend/fix/main-test-fail #121

Re-run all jobs ⋮

Summary

All jobs

✓ backend-test

✓ backend-build-and-deploy

Run details

Usage

Workflow file

Triggered via push 13 minutes ago

AnthonyChing pushed 492bfb4 v1.1.2

Status

Success

Total duration

5m 31s

Artifacts

1

backend.yml

on: push

✓ backend-test 2m 42s

✓ backend-build-and-de... 2m 42s

backend-build-and-deploy summary

Docker Build summary

For a detailed look at the build, download the following build record archive and import it into Docker Desktop's Builds view. Build records include details such as timing, dependencies, results, logs, traces, and other information about a build. [Learn more](#)

AnthonyChing~performance-management-system~MHOXMZ.dockerbuild (373.48 KB - includes 1 build record)

Find this useful? [Let us know](#)

ID	Name	Status	Cached	Duration
MHOXMZ	backend	✓ completed	0%	1m10s

► Build inputs

SonarCloud Report

Overview

No tags · 10k Lines of Code ⓘ · Last analysis 30 minutes ago

Quality Gate Status



Passed
All conditions passed

Open Issues

Overall code · Status: Open

65

Duplications

Overall code

2.1%

↘ -76.7% vs last 30 days

Coverage

Overall code

83.7%

↗ +56.2% vs last 30 days

Security snapshot

Security Rating

Overall code



Security Issues

A

Overall code · Software quali...

0

↘ -100.0% vs last 30 days

Open Security Issues by Severity

Overall code · Software quality: Security · Slice by: Severity

No data available to display

前端單元測試與 Playwright E2E Test

- 使用 Jest / Vitest 確保 UI 元件與 API 串接正確。
- 使用 Playwright 撰寫 E2E Test，由機器人自動模擬員工、主管、HR三個角色的實際點擊與操作流程，確保核心業務邏輯不中斷。

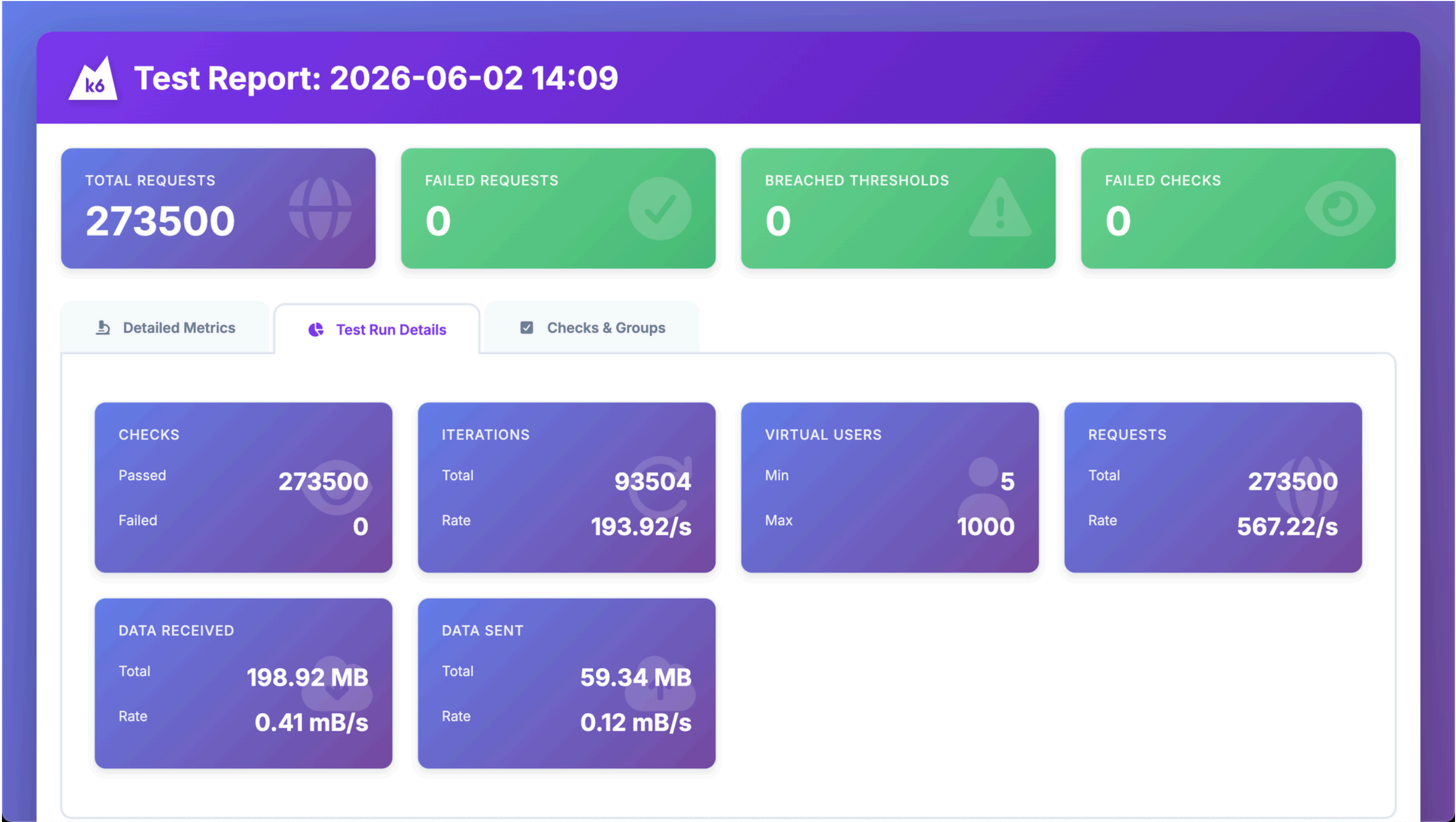
Integration Testing with Testcontainers

- **Testcontainers + Docker**：在測試時，程式會自動啟動一個真實的 PostgreSQL 容器，確保所有的 SQL 語法與資料庫交易在開發環境的測試結果，與 production 環境 100% 一致。

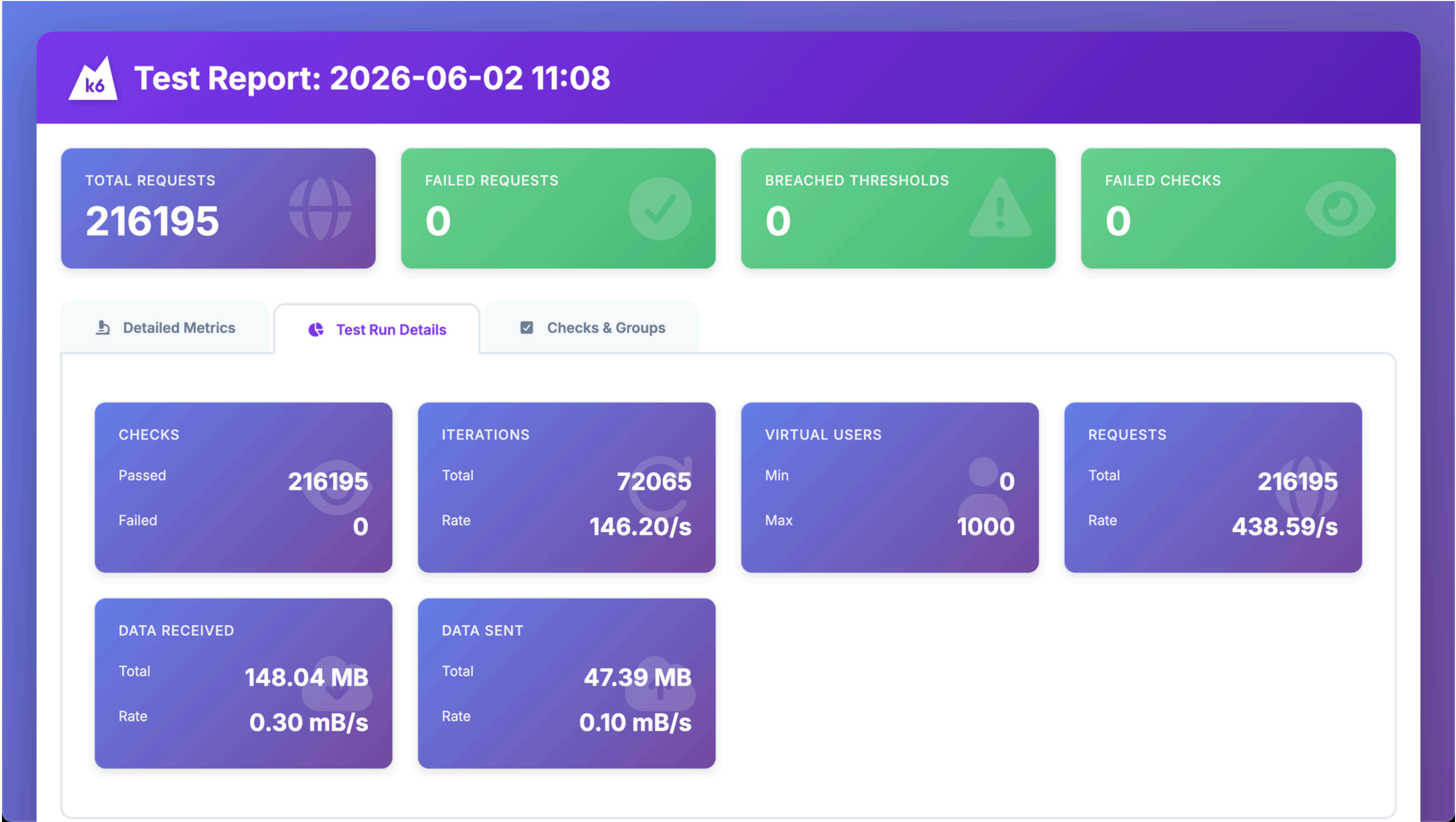
k6 Test 策略

- **Little's Law:** $L = \lambda \times W$ (系統內人數 = 到達率 $\times$ 停留時間)
 - $L = (30,000 \text{ 人} \times 6 \text{ 分鐘}) \div 60 \text{ 分鐘} = 3,000 \text{ CCU}$
 - 突發係數 (Burst Factor) : 1.75x
 - **Ultimate CCU** = $3,000 \times 1.75 \approx \mathbf{5,000 \text{ CCU}}$
- **3 Test Scenarios:**
 - **Load Test:** 正常尖峰時段的承載能力與穩定度。
 - **Spike Test:** 模擬 HR 寄出群發信後，系統面對瞬間海量登入的抗壓性。
 - **Stress Test:** 主動尋找系統的崩潰臨界點。

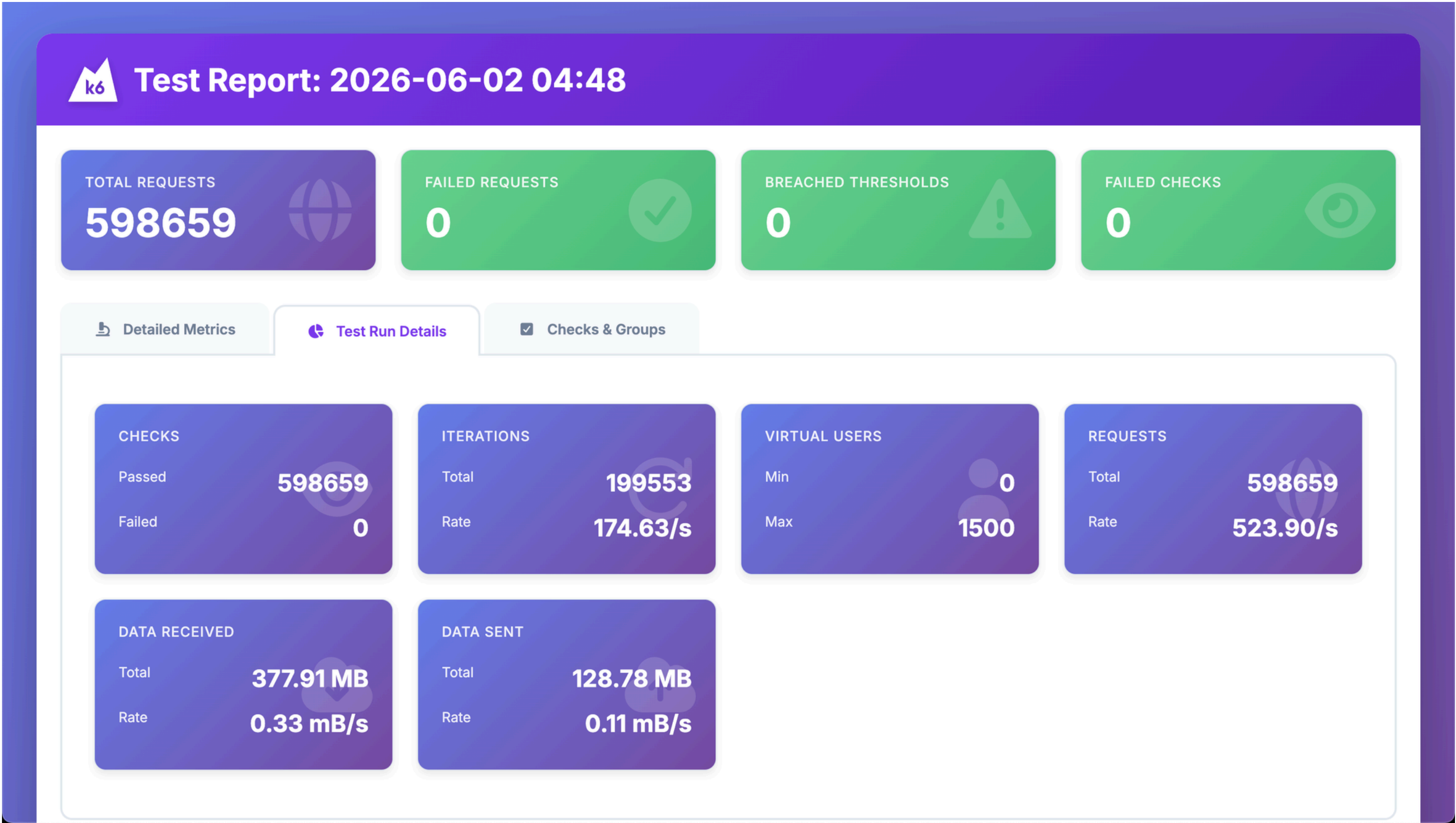
k6 Load Test 結果



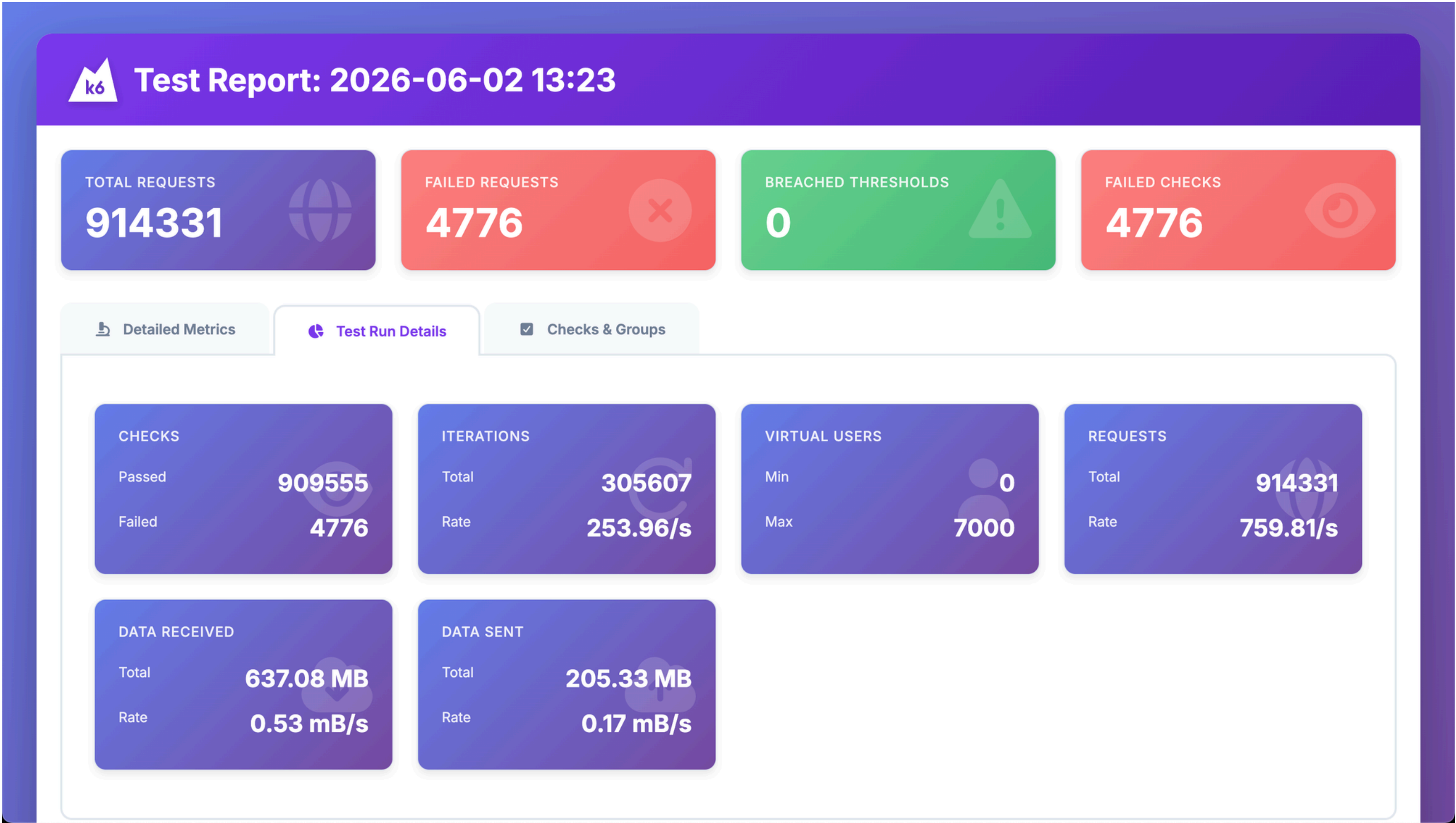
k6 Spike Test 結果



k6 Stress Test 結果 (加壓到 1500 VUs)



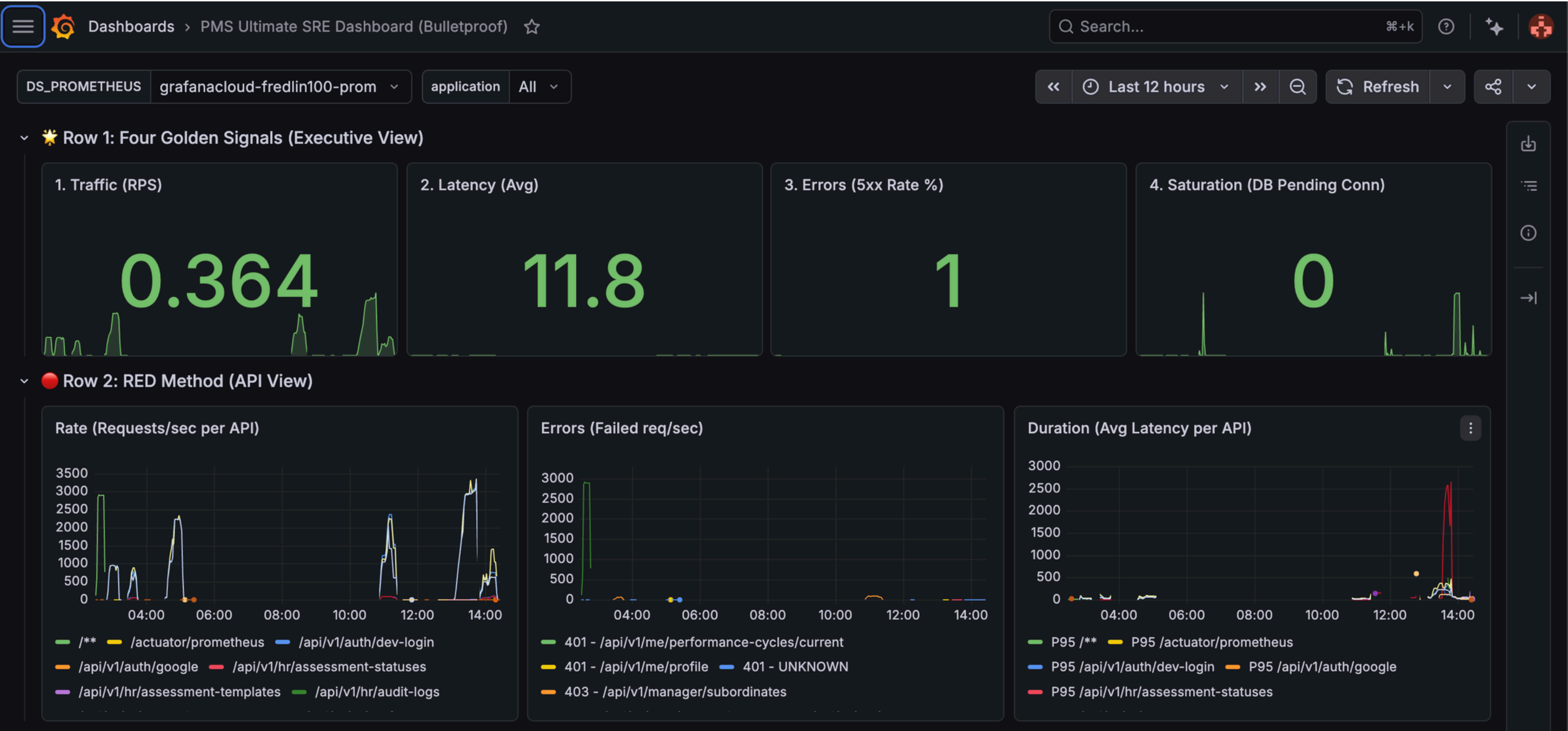
k6 Stress Test 結果 (加壓到 7000 VUs)

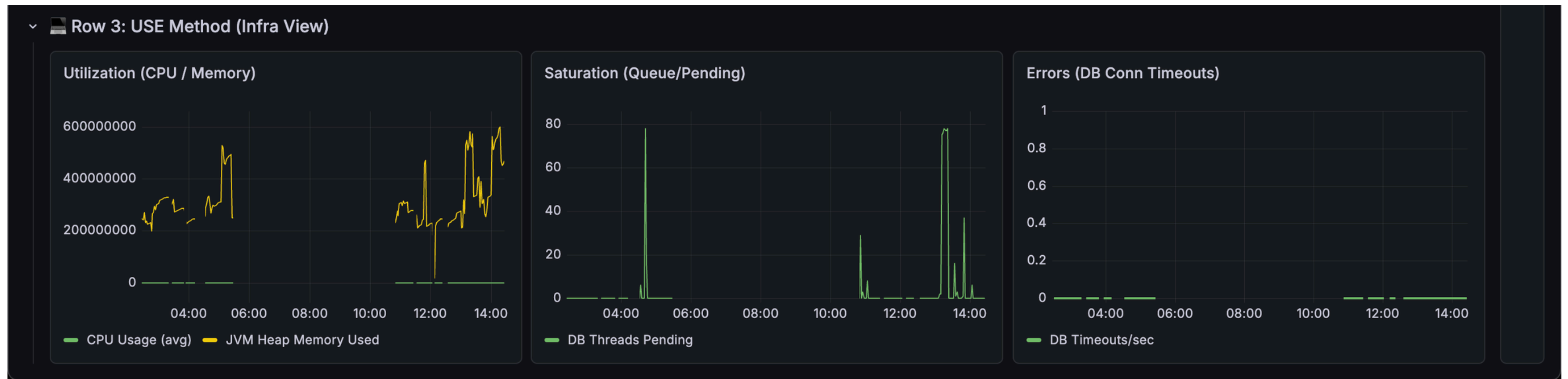


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Grafana





融合 Four Golden Signals、RED Method、USE Method 的監控架構

Four Golden Signals

- **Traffic**
每秒 HTTP 請求數
- **Latency**
處理請求的平均時間
- **Errors**
HTTP 5XX rate
- **Saturation**
DB Pending Connections

RED Method

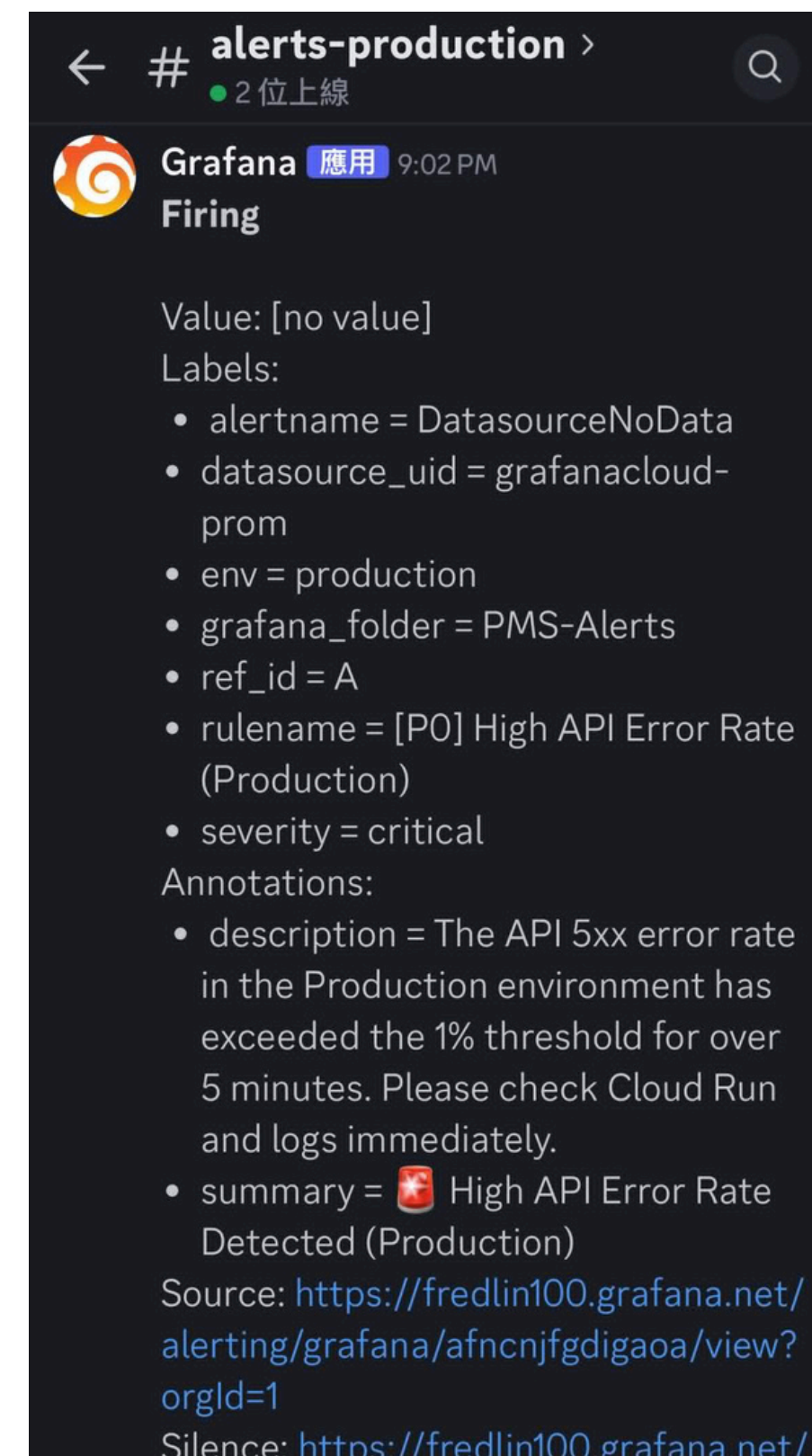
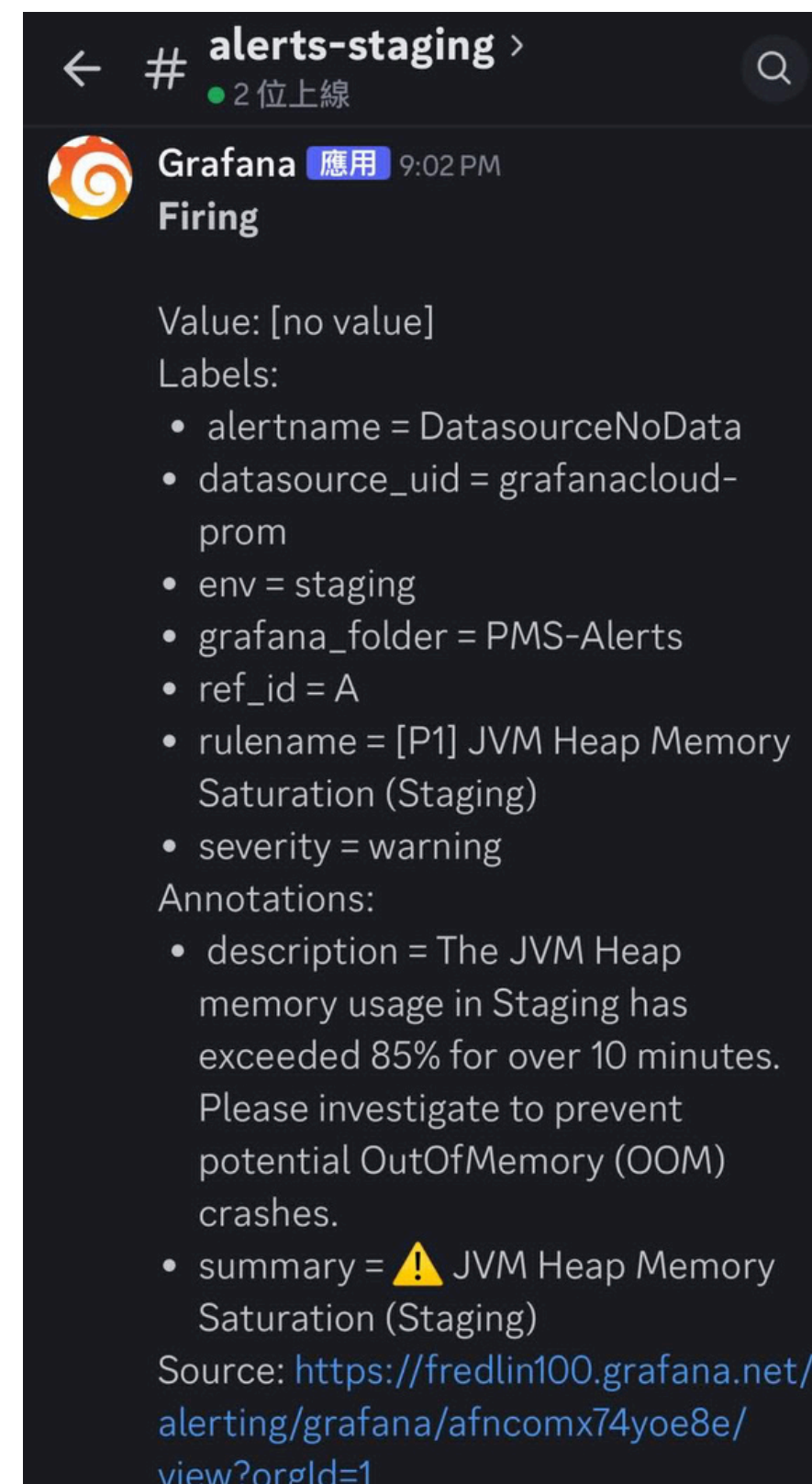
- **Request Rate**
計算各 API RPS
- **Request Errors**
列出 4xx / 5xx API
- **Request Duration**
各 API 平均處理延遲

USE Method

- **Utilization**
CPU Usage / JVM
Memory Used
- **Saturation**
DB Threads Pending
- **Errors**
DB 連線超時錯誤

整合 Discord 達到告警標準即發送 Alert 訊息

- **HTTP 5xx 錯誤率：**
最近 1 分鐘內，HTTP 5xx 佔所有請求百分比 > 1%，持續5分鐘
- **P95 API 處理延遲：**
95th Percentile 延遲 > 1分鐘
持續5分鐘
- **JVM Heap Memory 使用率：**
已使用的 Heap ÷ 最大可用的 Heap > 85%，持續10分鐘



高可用性與高安全性

- **高可用性**：將 GCP 服務由 單一可用區 切換至 多可用區
- **高安全性**：
 - 將「員工績效評分」與「薪資/個資」等敏感欄位，在寫入資料庫前使用 AES 演算法進行加密。
 - 採用 GCM 模式確保隨機初始化向量、保證資料完整性
 - 整合 Google Workspace (OAuth2 / OIDC) 以達到支援 MFA 登入



THANKS!